Mini Project Report

Air Quality Index (AQI) Monitoring

Submitted By

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CSE Department | Section S5

R-Programming Lab

SCSVMV UNIVERSITY

Experiment 3: Air Quality Index (AQI) Monitoring

Aim

To collect, analyse, and visualize Air Quality Index (AQI) data from multiple cities using R programming, and to interpret pollution levels using bar charts, histograms, box plots, and pie charts.

Scenario / Background

- Environmental monitoring agencies collect daily AQI readings from sensors placed across major Indian cities.
- The AQI value reflects the concentration of key pollutants: PM2.5, PM10, CO, NO2, SO2, and O3.
- Government bodies and the public use AQI categories (Good, Moderate, Unhealthy, Very Unhealthy, Hazardous) to decide health advisories.
- Analysing AQI trends helps identify cities with chronic pollution problems and the most common primary pollutants.
- Visualizing the data allows policymakers to take targeted action to improve air quality and public health outcomes.

Sample Dataset

The dataset contains 10 records from 5 Indian cities collected over 10 consecutive days.

Station ID	Date	City	AQI Value	Pollutant Conc. (µg/m³)	Primary Pollutant	AQI Category
S001	2024-01-15	Chennai	142	45	PM2.5	Unhealthy
S002	2024-01-16	Mumbai	89	30	PM10	Moderate
S003	2024-01-17	Delhi	210	60	PM2.5	Very Unhealthy
S004	2024-01-18	Kolkata	55	20	CO	Good
S005	2024-01-19	Bangalore	78	25	NO2	Moderate
S006	2024-01-20	Chennai	165	50	PM2.5	Unhealthy
S007	2024-01-21	Mumbai	38	10	SO2	Good
S008	2024-01-22	Delhi	295	80	PM2.5	Hazardous
S009	2024-01-23	Kolkata	110	35	PM10	Unhealthy for Sensitive
S010	2024-01-24	Bangalore	62	22	O3	Moderate

R Program (with line-by-line comments)

```
# ============================================
# Experiment 3: Air Quality Index Monitoring
# ============================================

# Step 1: Load required libraries
library(ggplot2)   # For creating visualizations
library(dplyr)     # For data manipulation

# Step 2: Define the AQI dataset inline
aqi_data <- data.frame(
  StationID = c("S001","S002","S003","S004","S005",
"S006","S007","S008","S009","S010"),
  Date      = as.Date(c("2024-01-15","2024-01-16","2024-01-17",
"2024-01-18","2024-01-19","2024-01-20",
"2024-01-21","2024-01-22","2024-01-23",
"2024-01-24")),
  City      = c("Chennai","Mumbai","Delhi","Kolkata","Bangalore",
"Chennai","Mumbai","Delhi","Kolkata","Bangalore"),
  AQI       = c(142, 89, 210, 55, 78, 165, 38, 295, 110, 62),
  Pollutant_Conc = c(45, 30, 60, 20, 25, 50, 10, 80, 35, 22),
  Primary_Pollutant = c("PM2.5","PM10","PM2.5","CO","NO2",
"PM2.5","SO2","PM2.5","PM10","O3"),
  Category  = c("Unhealthy","Moderate","Very Unhealthy","Good",
"Moderate","Unhealthy","Good","Hazardous",
"Unhealthy for Sensitive","Moderate")
)

# Step 3: View the dataset
View(aqi_data)
cat("AQI Dataset Loaded Successfully\n")
print(summary(aqi_data))

# Step 4: Bar Chart - Average AQI by City
avg_aqi <- aqi_data %>%
  group_by(City) %>%
  summarise(Avg_AQI = mean(AQI))   # Compute mean AQI per city

ggplot(avg_aqi, aes(x = City, y = Avg_AQI, fill = City)) +
  geom_bar(stat = "identity") +    # Draw bars with actual values
  labs(title = "Average AQI by City",
       x = "City", y = "Average AQI") +
  theme_minimal() +
  theme(legend.position = "none")  # Remove legend (redundant with x-axis)

# Step 5: Histogram - AQI Value Distribution
ggplot(aqi_data, aes(x = AQI)) +
  geom_histogram(binwidth = 40, fill = "steelblue", color = "black") +
  labs(title = "Distribution of AQI Values",
       x = "AQI", y = "Frequency") +
  theme_minimal()

# Step 6: Box Plot - AQI spread per City
ggplot(aqi_data, aes(x = City, y = AQI, fill = City)) +
  geom_boxplot() +                 # Show median, IQR, and outliers
  labs(title = "AQI Box Plot by City",
       x = "City", y = "AQI") +
  theme_minimal() +
  theme(legend.position = "none")
```

```
# Step 7: Pie Chart - Proportion of AQI Categories
category_counts <- table(aqi_data$Category)   # Count each category
pie(category_counts,
    main = "AQI Category Distribution",
    col  = rainbow(length(category_counts)))  # Assign distinct colors
```

Output / Observation

- Bar Chart: Delhi records the highest average AQI (~252), followed by Chennai (~153), indicating severe pollution levels. Bangalore and Kolkata show relatively lower average AQI values.
- Histogram: The AQI distribution is right-skewed, with most readings clustered between 38 and 165. One extreme outlier (AQI = 295 for Delhi) highlights hazardous episodes.
- Box Plot: Delhi shows the widest interquartile range and highest median AQI, confirming persistent air quality issues. Mumbai and Bangalore have compact, lower-range distributions.
- Pie Chart: PM2.5 is the dominant primary pollutant (~40% of records), followed by PM10 and other pollutants, reflecting the major role of particulate matter in urban air pollution.
- Overall Insight: The analysis reveals that metro cities like Delhi face chronic high-AQI conditions, primarily driven by PM2.5. Smaller cities like Bangalore maintain relatively better air quality. Monitoring such trends is critical for data-driven environmental policymaking.

Graphs Representation

The following graphs are generated by executing the above R program:

- Graph 1: Bar Chart – Average AQI by City
- Graph 2: Histogram – Distribution of AQI Values
- Graph 3: Box Plot – AQI Spread per City
- Graph 4: Pie Chart – AQI Category Distribution

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Commands + Code + Text Run all

[1]

```
# ==========================================
# Experiment 3: Air Quality Index Monitoring
# ==========================================

# Step 1: Load required libraries
library(ggplot2)   # For graphs
library(dplyr)     # For data handling

# Step 2: Create AQI dataset
aqi_data <- data.frame(
  StationID = c("S001","S002","S003","S004","S005",
                "S006","S007","S008","S009","S010"),

  Date = as.Date(c("2024-01-15","2024-01-16","2024-01-17",
                   "2024-01-18","2024-01-19","2024-01-20",
                   "2024-01-21","2024-01-22","2024-01-23",
                   "2024-01-24")),

  City = c("Chennai","Mumbai","Delhi","Kolkata","Bangalore",
           "Chennai","Mumbai","Delhi","Kolkata","Bangalore"),

  AQI = c(142,89,210,55,78,165,38,295,110,62),

  Pollutant_Conc = c(45,30,60,20,25,50,10,80,35,22),

  Primary_Pollutant = c("PM2.5","PM10","PM2.5","CO","NO2",
                        "PM2.5","SO2","PM2.5","PM10","O3"),

  Category = c("Unhealthy","Moderate","Very Unhealthy","Good",
               "Moderate","Unhealthy","Good","Hazardous",
               "Unhealthy for Sensitive","Moderate")
)

# Step 3: View dataset
View(aqi_data)
cat("AQI Dataset Loaded Successfully\n")
print(summary(aqi_data))

# Step 4: Bar Chart - Average AQI by City
avg_aqi <- aqi_data %>%
  group_by(City) %>%
```

Variables Terminal R

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```
  summarise(Avg_AQI = mean(AQI))

ggplot(avg_aqi, aes(x = City, y = Avg_AQI, fill = City)) +
  geom_bar(stat = "identity") +
  labs(title = "Average AQI by City",
       x = "City",
       y = "Average AQI") +
  theme_minimal() +
  theme(legend.position = "none")

# Step 5: Histogram - AQI Value Distribution
ggplot(aqi_data, aes(x = AQI)) +
  geom_histogram(binwidth = 40, fill = "steelblue", color = "black") +
  labs(title = "Distribution of AQI Values",
       x = "AQI",
       y = "Frequency") +
  theme_minimal()

# Step 6: Box Plot - AQI Spread by City
ggplot(aqi_data, aes(x = City, y = AQI, fill = City)) +
  geom_boxplot() +
  labs(title = "AQI Box Plot by City",
       x = "City",
       y = "AQI") +
  theme_minimal() +
  theme(legend.position = "none")

# Step 7: Pie Chart - AQI Category Distribution
category_counts <- table(aqi_data$Category)

pie(category_counts,
    main = "AQI Category Distribution",
    col = rainbow(length(category_counts)))
```

```
Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag
```

Variables Terminal R

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```
col = rainbow(length(category_counts)))
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

A data.frame: 10 × 7

StationID	Date	City	AQI	Pollutant_Conc	Primary_Pollutant	Category
<chr>	<date>	<chr>	<dbl>	<dbl>	<chr>	<chr>
S001	2024-01-15	Chennai	142	45	PM2.5	Unhealthy
S002	2024-01-16	Mumbai	89	30	PM10	Moderate
S003	2024-01-17	Delhi	210	60	PM2.5	Very Unhealthy
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S005	2024-01-19	Bangalore	78	25	NO2	Moderate
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S007	2024-01-21	Mumbai	38	10	SO2	Good
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S009	2024-01-23	Kolkata	110	35	PM10	Unhealthy for Sensitive
S010	2024-01-24	Bangalore	62	22	O3	Moderate

```
AQI Dataset Loaded Successfully
  StationID              Date                City               AQI
 Length:10          Min.   :2024-01-15   Length:10          Min.   : 38.0
 Class :character   1st Qu.:2024-01-17   Class :character   1st Qu.: 66.0
 Mode  :character   Median :2024-01-19   Mode  :character   Median : 99.5
                    Mean   :2024-01-19                      Mean   :124.4
                    3rd Qu.:2024-01-21                      3rd Qu.:159.2
                    Max.   :2024-01-24                      Max.   :295.0
 Pollutant_Conc  Primary_Pollutant    Category
 Min.   :10.00   Length:10          Length:10
 1st Qu.:22.75   Class :character   Class :character
 Median :32.50   Mode  :character   Mode  :character
 Mean   :37.70
 3rd Qu.:48.75
 Max.   :80.00
```

Average AQI by City

Variables Terminal

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11249A002.ipynb

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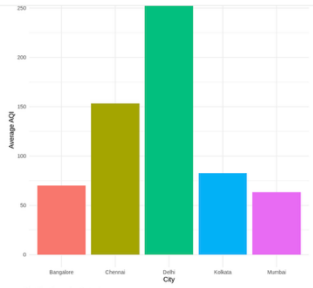
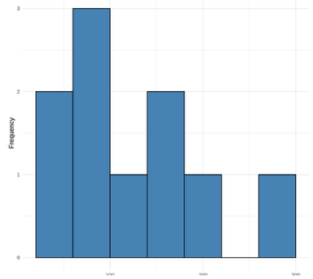
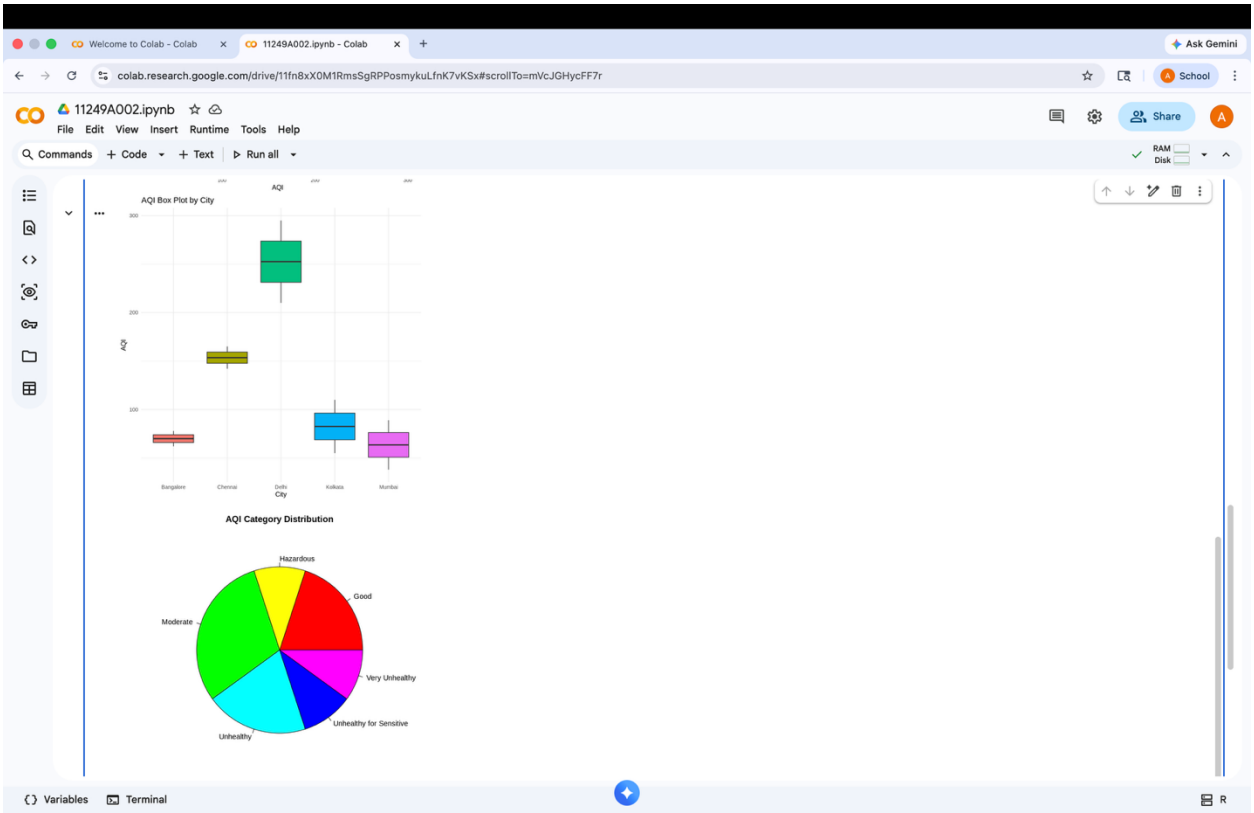
```
col = rainbow(length(category_counts)))
```



Distribution of AQI Values



Variables Terminal



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